Quantum Physics

Questions

1. Explain Heisenberg's uncertainty principle.
2. Give the physical significance of wave function.
3. Explain the de-Broglie hypothesis.
4. Calculate the de-Broglie wavelength of 5KeV neutron. Given the mass of the neutron is 1.675×10^{-27} kg.
5. What are matter waves? Obtain an expression for the wavelength of matter waves.
6. Explain the difference between a matter wave and an electromagnetic wave.
7. Explain the properties of matter waves.
8. Derive time-independent Schrodinger's wave equation for a free particle.
9. Show that the wavelength λ associated with an electron of mass 'm' and kinetic energy 'E' is given by $\lambda = h/\sqrt{2mE}$.
10. Define the Photoelectric effect.
11. Find the de-Broglie wavelength of a neutron of energy 12.8MeV.
12. Explain the duality of matter waves from the inferences drawn from the photoelectric effect.
13. Describe the theory of one-dimensional particles in a box.
14. Derive an expression for Schrodinger's time-independent wave equation. Explain the significance of a wave function.
15. Explain the concept of wave-particle duality and obtain an expression for the wavelength of matter waves.
16. Derive eigen values and eigen functions for a particle in a one-dimensional box.
17. What is the Photoelectric effect? Explain the Photoelectric effect and obtain its Work Function.
18. State and explain Compton Effect.
19. Define wave function. What is meant by normalized wave function? Explain
20. What are the failures of classical physics? What phenomena led to the discovery of quantum physics?
21. Give a few phenomena where waves behave like particles. Give a few phenomena where particles behave like waves.
22. Give Einstein's theory of the photoelectric effect.
23. What do you mean by the work function of a metal? Derive an expression for it.
24. What do you mean by stopping the potential of a metal? Derive an expression for it.
25. What do you mean by the threshold frequency of a metal? Derive an expression for it.
26. What is the Compton effect? Explain how the classical concept failed in explaining the Compton effect.
27. Explain why the frequency of the scattered photon is less than that of the incident photon in the case of Compton scattering.
28. What is Compton's shift? What are the factors on which the Compton shift depends?
29. What is the de Broglie hypothesis?

30. What is the de Broglie wave? Derive an expression for the wavelength of the de Broglie waves of a material particle moving with relativistic speed.
31. What is the de Broglie relation for a charged particle moving under a potential difference of V volts?
32. State and explain Heisenberg's uncertainty principle.
33. What is a wave function? What are the characteristics of a wave function?
34. What is probability density? Derive an expression for it.
35. Derive an expression for the probability of finding a particle described by the wave function ψ in a certain region.
36. What are the units of one, two, and three-dimensional wave functions?
37. State and explain the superposition principle for wave functions.
38. What do you mean by the normalization of a wave function?
39. What is the meaning of a normalization constant?
40. How would you find the normalization constant of a wave function?
41. Give a few observables and operators of quantum mechanics.
42. What do you mean by eigenvalues of a physical quantity? Explain.
43. What do you mean by the eigenfunctions of a physical system? Explain.
44. Calculate the momentum eigenfunctions and momentum eigenvalues of a particle trapped in a one-dimensional box of length L when the particle is described by the normalized wave function

    $$\psi = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}.$$

    Also, calculate the energy eigenvalues of the particle.
45. What do you mean by the expectation value of a physical quantity? What does it represent?
46. The expectation value of the momentum of a particle trapped in a one-dimensional box of length L described by the normalized wave function $\psi = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}$ is zero. Explain.
47. What is the time-independent Schrödinger's equation for a particle of mass m and energy E moving along the (i)X-axis, (ii) Y –axis, and (iii) Z –axis under a potential energy V?
48. What is the time-dependent Schrödinger's equation for a particle of mass m and energy E moving along the (i)X-axis, (ii) Y –axis, and (iii) Z –axis under a potential energy V?
49. Find the eigenfunction for the operator $i\hbar$ d/dx.
50. What is the condition for a particle to be a free particle?
51. Solve Schrödinger's time-independent equation for a free particle.
52. What are the boundary conditions on y so that it turns out a physically meaningful wave function?
53. What do you mean by a free particle? Derive the expression for its wave function.

54. Derive an expression for the normalized wave function of a particle trapped in an infinitely deep potential well.
55. What are the eigenfunctions of a particle trapped in an infinite deep potential well in its ground state and third excited state?
56. Prove that the energy of the particle enclosed in an infinite deep potential well is quantized.
57. Prove that the momentum of a particle in a one-dimensional potential well of infinity height is quantized.
58. Derive an expression for the energy of the particle enclosed in an infinite deep potential well.
59. Explain how quantum physics differs from classical physics in infinite deep potential well problems.
60. A particle of mass m is enclosed inside a potential well of infinite height. What is the lowest speed the particle is to be given so that the particle acquires sustained states of motion?
61. Prove that the spacing between consecutive energy levels of a particle inside an infinite deep potential well increases with the increase of the energy.
62. a) Calculate the minimum uncertainty in the position of (i) a ball of mass 150 g and (ii) an electron if their velocity can be measured to a precision of 0.50×10^{-3} ms^{-1}.

    b) Calculate the minimum uncertainty in the momentum of a ^4He atom confined to a 0.40 nm region.
63. Calculate the minimum energy of a particle of mass m confined in a one-dimensional box of length a. Using these results estimate the minimum kinetic energy and the corresponding speed for a particle of mass 1.0×10^{-8} kg confined to a region of 1.0×10^{-6} m.
64. Suppose the position of an electron can be measured in an atom to within an accuracy of 0.2 Å. What would be the uncertainty in its velocity? What would be its minimum kinetic energy?
65. An atom in an excited state decays by the emission of a photon. If the average lifetime of an excited atom is ~ 2.0×10^{-8} s, what is the minimum uncertainty in the energy of the emitted photon? What will be the order of the natural width ($\Delta\nu$) of the spectral line of the atomic emission?
66. a) The average decay lifetime of an elementary particle is 2.0 μs. Calculate the minimum uncertainty in the measurement of its energy. Also, calculate the minimum uncertainty in the measurement of its mass using the mass-energy equivalence.

    b) The uncertainty in the energy of a short-lived particle is measured to be 0.40MeV. What is the smallest lifetime the particle can have?
67. Calculate the minimum energy for an electron in an atom given that the typical size of an atom is ~ 1Å.

68. Calculate the de Broglie wavelength for a cricket ball traveling at a speed of $50\ ms^{-1}$, given that its mass is 150g.
69. Calculate the de Broglie wavelength of an electron accelerated from rest through a potential difference of 100 V.
70. The particle trapped in a one-dimensional box of length L is described by the normalized wave function

    $$\psi = \sqrt{\frac{2}{L}} \sin\frac{n\pi x}{L}.$$

    What is the probability that the particle in the nth state is lying between 0 and L/n?
71. If 1/2, 1/3, and 1/6 are the probabilities that the system be in the three states represented by eigenfunctions ψ_1, ψ_2, and ψ_3, respectively, what is the wave function of the system?
72. 2/5, 1/5, 3/10, and 1/10 are the probabilities that the system be in the four states having energy eigenvalues 4 eV, 6 eV, 7 eV, and 9 eV, respectively. What is the energy expectation value of the system?
73. The wave function of a system is given by $\psi = p\psi_1 + q\psi_2 + r\psi_3$. What are the probabilities of finding the system in the states ψ_1, ψ_2, and ψ_3?
74. a) The stopping potential observed in a certain experiment on the photoelectric effect is 5.0 V. Calculate the maximum kinetic energy and the maximum speed of the photoelectrons.

    b) The cut-off wavelength for copper is 296 nm. Calculate the cut-off frequency, the work function, and the maximum kinetic energy of the photoelectrons when ultraviolet light of wavelength 250 nm is incident on a copper photoelectrode.

75. Violet light of wavelength 500 nm is incident on a photoelectrode having a work function of 2.28 eV. Calculate the maximum kinetic energy of ejected electrons and the stopping potential.

76. X-rays of wavelength 50.0 pm are incident on a target. Calculate the wavelength of the scattered X-rays at the angles of (i) 45˚ and (ii) 60˚.

77. If the Compton shift is equal to one Compton wavelength, what is the scattering angle? What is the value of θ for which the value of Δλ is maximum? What is this value of Δλ?

78. A particle of mass m can move freely along the x-axis between $-L \leq x \leq L$ but is prohibited from going outside this region. Determine the value of the constant E for which the following wave function is a solution of the time-dependent Schrödinger equation for the particle:

$$\psi(x,t) = \begin{cases} N\sin\left(\frac{\pi x}{L}\right) Exp(-\frac{i2\pi Et}{h}), & -L < x < L \\ 0, & Everywhere \end{cases}$$

79. If ψ_1(x.t) and ψ_2 (x,t) are two solutions of the Schrödinger equation, show that a linear combination of ψ_1 and ψ_2, $\psi = a_1\psi_1 + a_2\psi_2$ with a_1 and a_2 as arbitrary constants, is also a solution of the same Schrödinger equation.

80. Express the variables p_y and p_z in operator form.

81. Show that the operators $\hat{x}$ and $\hat{p}_x$ are linear operators.

82. Calculate <x> for the wave function:

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right), 0 < x < L$$

83. The normalized wave function of a particle moving in the x-direction is given by:

$$\psi(x) = \sqrt{\frac{30}{a^5}}\, x(a-x), \qquad 0 < x < a, \;\; Calculate < x >.$$

84. Calculate the normalization constant N for the stationary state wave function for a system given by:

$$\psi(x) = N \exp\left(-\frac{x^2}{2}\right) \qquad \text{for } -\infty \le x \le \infty$$

85. At t =0, the wave function of a particle is given by:

$$\psi(x,t) = \begin{cases} Ax(a-x) & \text{for } 0 \le x \le a \\ 0 & \text{elsewhere} \end{cases}$$

Calculate the probability of finding the particle between x = 0 and x = a / 2.

86. Show that $\psi(x) = Ae^{ikx} + Be^{-ikx}$ is an eigenfunction of the time-independent Schrödinger equation for the quantum mechanical free particle with an eigenvalue $E = \hbar^2 k^2/2m$.

87. Determine the normalization constant A for the following eigenfunction of a particle confined in a box of length L: $\Psi n(x) = A \sin k_n x = A \sin(n\pi x/L)$, $0 < x < L$.

88. Show that the eigenfunctions for the particle in a box are orthogonal.

89. A toy car of mass 0.2 kg is moving back and forth on a track 1.0 m long. Treating the car as a particle in a box, what would be the minimum energy of this car? Suppose it moves with a constant speed of $0.1 ms^{-1}$; what is the value of n corresponding to this speed?

90. Calculate the energy levels of an electron confined to an atomic box of size 1.0 Å.

91. Calculate the probability of finding the particle between

(i) $x = L/4$, and $x = 3L/4$ and

(ii) $x = 0$ and $x = L/4$

when the particle is in the ground state

92. Calculate the probability of finding the particle between $x = L / 4$ and $x = 3L/ 4$ when the particle is in the first excited state.